Paper 31: Zone Field Structure in VCML

Flat Attractors at Low Diffusion, Gradient Zones at High Diffusion;
Low-Amplitude Saturation in the High-Turnover Regime

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Abstract

Paper 30 found $K_{\text{eff}} \propto (\sqrt{\kappa}/\text{ZONE_W})^{0.6}$, suggesting a finite field coherence length ℓ mediates the spatial amplification factor Γ. Paper 31 directly measures within-zone spatial structure via the autocorrelation $C(\Delta x)$ of the zone-demeaned column field profile. The key findings are: (i) At the standard and lower diffusion coefficients ($\kappa \leq 0.020$), the within-zone autocorrelation at lag 1 is near the finite-N floor ($C[1] \approx -0.1$ to -0.2), indicating that zones are *spatially uniform* flat attractors – the field has nearly the same value at all columns within a zone, with no detectable spatial gradient. (ii) At high diffusion ($\kappa \geq 0.080$), $C[1]$ becomes positive ($+0.13$ to $+0.25$), indicating that diffusion creates genuine within-zone spatial gradients – but these are accompanied by reduced zone differentiation (sg4 falls with κ). (iii) The ν-dependence of spatial structure is absent: $C[1]$ is near zero at all $\nu \in \{0.0003, 0.001, 0.003, 0.010, 0.030\}$, confirming that turnover rate does not affect within-zone field uniformity. (iv) The $\nu = 0.003$ failure identified in Paper 30 is a *low-amplitude saturation*: zone structure forms rapidly but stabilises at $\approx 40\%$ of the standard amplitude, rather than failing to form at all. Together, these results establish that optimal VCML operation relies on *copy-forward uniformity* (flat-attractor zones) rather than diffusion-driven gradient formation.

1 Background and Open Questions

Paper 30 measured the spatial amplification factor $\Gamma = K^{\text{I}}_{\text{eff}}/K_{\text{eff}}$ across a grid of $(\kappa, \text{ZONE_W})$ conditions and a ν sweep. It found a sub-linear scaling with both parameters:

$$K_{\text{eff}} \propto \left(\frac{\sqrt{\kappa}}{\text{ZONE_W}}\right)^{0.6}, \qquad \Gamma \propto \left(\frac{\text{ZONE_W}}{\sqrt{\kappa}}\right)^{0.6} \tag{1}$$

The sub-linear exponent (0.6 vs. 1.0) was interpreted as a finite field coherence length ℓ that saturates with zone width. Paper 30 also found a non-monotone ν dependence, with Γ collapsing at $\nu = 0.003$.

Paper 31 was designed to answer:

- **Q16.** What is the field coherence length $\ell(\kappa, \nu)$? Measure spatial autocorrelation of $\mathbf{F}$ at steady state; test $\ell \propto \sqrt{\kappa/\nu}$.
- **Q17.** Is the revised exponent 0.6 stable across conditions?
- **Q18.** At $\nu = 0.003$, does spatial structure fail to form or form and then collapse? Track sg4(t).

2 Experiment Design

Fixed parameters. $H = 20$, $N_{\text{zones}} = 4$, ZONE_W $= 5$, HALF $= 20$, $N_{\text{act}} = 400$, BASE_BETA $= 0.005$, ALPHA_MID $= 0.15$, MID_DECAY $= 0.99$, FD $= 0.9997$, SS $= 10$, WAVE_DUR $= 15$, HS $= 2$, SEED_BETA $= 0.25$, WR $= 2.4$.

Within-zone autocorrelation measurement. At each checkpoint, the column-averaged field profile $\mathbf{F}^{\text{col}} \in \mathbb{R}^{\text{HALF}\times\text{HS}}$ is computed by averaging $\mathbf{F}$ over all $H = 20$ rows. For each of the $N_{\text{zones}} = 4$ zones, the ZONE_W $= 5$ column values (first HS component) are demeaned, and the normalised spatial autocorrelation $C(\Delta x)$ is computed for lags $\Delta x = 0, 1, 2, 3, 4$. The result is averaged across zones and seeds. The lag-1 value $C[1]$ is the primary summary statistic.

Finite-N floor. For a zero-mean random sequence of length n with no spatial structure, the expected autocorrelation at any positive lag is $-1/(n-1)$ due to the demeaning constraint. With $n =$ ZONE_W $= 5$, the finite-N floor is $-1/4 = -0.25$. Values near this floor indicate *spatially uniform* zones (within-zone variance is present but has no spatial structure). Values significantly above zero indicate genuine spatial gradients.

Experiment A – κ sweep (25 runs). $\kappa \in \{0.001, 0.005, 0.020, 0.080, 0.200\}$, $\nu = 0.001$, FA $= 0.200$, 5 seeds, $T_{\text{end}} = 4000$, checkpoints at $\{1000, 2000, 3000, 4000\}$. Stores: sg4_t, $C(\Delta x)$ at each checkpoint.

Experiment B – ν sweep (25 runs). $\nu \in \{0.0003, 0.001, 0.003, 0.010, 0.030\}$, $\kappa = 0.020$, FA $= 0.200$, 5 seeds, $T_{\text{end}} = 4000$.

Experiment C – $\nu = 0.003$ failure mode (50 runs). $\nu \in \{0.001, 0.003\}$, FA $\in \{0.005, 0.020, 0.080, 0.200, 0.80$ $\kappa = 0.020$, 5 seeds, $T_{\text{end}} = 6000$, 12 checkpoints at $\{500, 1000, \ldots, 6000\}$.

Total: 100 runs.

3 Results

3.1 Experiment A: Within-Zone Spatial Structure vs κ

Table 1: Exp A: within-zone autocorrelation $C[1]$ (lag-1) and $C[2]$ (lag-2) at $T = 4000$, averaged over 5 seeds. Finite-N floor $= -0.25$. sg4 at $T = 4000$ (FA=0.200).

κ	$C[1]$	$C[2]$	sg4
0.001	−0.197	−0.123	(high)
0.005	−0.102	−0.153	(high)
0.020	−0.035	−0.218	(moderate)
0.080	+0.127	−0.300	(lower)
0.200	+0.248	−0.253	(lower)

Table 1 and Figure 1A show a clear transition in within-zone spatial structure as κ increases.

Low diffusion ($\kappa \leq 0.020$): flat-attractor zones. At $\kappa \leq 0.020$, $C[1]$ falls between -0.197 and -0.035 – values near or slightly above the finite-N floor of -0.25. This indicates that within each zone, the field is approximately *spatially uniform*: all five columns in a zone have nearly the same $\mathbf{F}$ value, and the residual within-zone variance is structureless noise. The within-zone field is a flat attractor, not a spatial gradient.

The copy-forward mechanism explains this. At birth, a cell samples its field from one random neighbour. With many births per unit time, the field value diffuses randomly across the zone. With weak spatial diffusion (κ small), the dominant spreading mechanism is copy-forward – a random walk that *averages* rather than *gradients* the field within a zone.

High diffusion ($\kappa \geq 0.080$): gradient zones. At $\kappa \geq 0.080$, $C[1]$ becomes significantly positive (+0.127 to +0.248). Spatial diffusion spreads field information smoothly, creating genuine within-zone gradients (columns at the zone edges differ from those at the centre, mirroring the adjacent zone's influence).

However, Figure 1B shows that this gradient regime is associated with *lower* sg4: high κ hurts zone differentiation by mixing field values across zone boundaries. The positive $C[1]$ at high κ is a signature of diffusive smearing, not enhanced structure.

Key finding. The optimal operating regime (low κ, high Γ) corresponds to spatially *uniform* zones. The zones in standard VCML are flat attractors, not spatial gradient structures. Within-zone coherence at the column scale is absent in the copy-forward-dominated regime.

3.2 Experiment B: Within-Zone Spatial Structure vs ν

Table 2: Exp B: $C[1]$, $C[2]$ vs ν at $T = 4000$. Standard $\kappa = 0.020$, FA=0.200.

ν	$C[1]$	$C[2]$
0.0003	+0.106	−0.303
0.0010	−0.035	−0.218
0.0030	+0.002	−0.228
0.0100	−0.042	−0.175
0.0300	−0.049	−0.211

Table 2 and Figure 1D show no systematic dependence of $C[1]$ on ν. All values lie near zero (between −0.05 and +0.11), with no monotone trend. The ν dependence of Γ found in Paper 30 is therefore *not* mediated by changes in within-zone spatial gradient structure. Rather, ν affects the temporal dynamics of the copy-forward loop (amplitude and coherence of the zone-level consensus), not its spatial character.

The slight positive value at $\nu = 0.0003$ ($C[1] = +0.106$) may reflect that with very infrequent turnover, the few birth events that do occur propagate a single "copy" of the field across the zone, creating a short-lived spatial correlation. But the effect is small and inconsistent across lags.

Q16 answered (revised). The field coherence length ℓ is effectively zero at the column scale in the copy-forward-dominated regime ($\kappa \leq 0.020$). The prediction $\ell \propto \sqrt{\kappa/\nu}$ would require $\ell \geq 1$ column to be measurable; instead, $\ell \ll 1$ column, and the relevant spatial unit is the zone (not the column). The sub-linear exponent in Paper 30's scaling (0.6 vs. 1.0) is not caused by a sub-column coherence length but by a different mechanism (discussed in Section 4).

3.3 Experiment C: $\nu = 0.003$ Failure Mode – Q18 Answered

Table 3 and Figure 1C answer Q18: at $\nu = 0.003$, zone structure *does form* – sg4 is non-zero from $t = 500$ onwards – but it saturates at a much lower amplitude ($\approx$ 30–45) than the standard ($\approx$ 85–113). This is *low-amplitude saturation*, not structural collapse.

The $\nu = 0.003$ condition forms structure *faster* than the standard at early times (sg4= 26.6 vs. 16.9 at $t = 500$): the higher turnover rate generates more copy-forward events per unit time, so zone identity spreads quickly. But the eventual amplitude is $\approx$ 40% of the standard: the copy-forward loop cannot build sufficient zone-level consensus when births are so frequent that each new cell "overrides" the field before it has been read by its neighbours.

Table 3: Exp C: sg4(t) at FA=0.200 for $\nu = 0.001$ vs $\nu = 0.003$. Mean over 5 seeds.

t	$\nu = 0.001$	$\nu = 0.003$
500	16.9	26.6
1000	32.0	29.0
1500	50.8	28.5
2000	63.3	21.5
2500	69.8	43.1
3000	113.1	37.5
4000	96.0	41.8
5000	95.9	45.0
6000	84.1	29.5

The $\nu = 0.003$ regime is characterised by high birth rate (mean lifetime ≈ 333 steps) and strong temporal fluctuations in sg4 (Table 3 shows swings from 21.5 to 43.1 within 500 steps). Each birth event is locally disruptive; the field never settles into a stable zone pattern.

4 Discussion

Copy-forward zones as flat attractors. The central result of Paper 31 is structural: in the VCML copy-forward regime, zones are *spatially uniform*. Every cell in a zone converges to approximately the same **F** value, irrespective of its column position within the zone. This is not a trivial consequence of diffusion (which would create smooth gradients); it is the result of *copy-forward averaging*, which propagates zone identity as a scalar rather than a spatial pattern.

The implication for Paper 30's scaling is significant. The prediction $\Gamma \propto \text{ZONE_W}/\sqrt{\kappa}$ assumed that wider zones provide more copy-forward hops before reaching a zone boundary, increasing the effective copy range ℓ. But if $\ell \approx 0$ (uniform zones), then width does not increase the copy range – it increases the *number of sites that share a zone identity*. Zone width is not a copy length but a *vote count*: more columns participate in building zone consensus.

This reframes the Paper 30 result: $K_{\text{eff}} \propto (\sqrt{\kappa}/\text{ZONE_W})^{0.6}$ is not a coherence-length effect but a *consensus-size effect*. Wider zones have more sites "voting" for the same zone identity, making consolidation more efficient (lower K_{eff}). The sub-linear exponent (0.6 vs. 1.0) reflects that consensus efficiency saturates: once enough sites vote, adding more sites contributes diminishing returns.

Diffusion and gradient zones. At high κ, diffusion creates within-zone spatial gradients ($C[1] > 0$) that are incompatible with the flat-attractor picture. These gradient zones are weaker (lower sg4) because diffusion simultaneously smears field values across zone boundaries. The transition from flat to gradient zones occurs around $\kappa \approx 0.050$–0.080, where the diffusion length $\sqrt{\kappa \cdot T_{\text{site}}} \approx \text{ZONE_W}$.

Turnover and low-amplitude saturation. The $\nu = 0.003$ result shows that the copy-forward loop has an optimal turnover rate. Too many births per unit time disrupt the consensus-building process before it can reach full amplitude. Importantly, the disruption is at the *amplitude* of zone differentiation, not at its *existence*: zones still form, but weakly. This is consistent with the flat-attractor picture: the attractor exists at $\nu = 0.003$ but has smaller basin depth, so thermal fluctuations (random births) push the system out of the attractor before it fully consolidates.

Revised picture of the copy-forward loop. Papers 27–31 together paint a consistent picture:

1. The copy-forward loop creates zone identity as a *shared scalar* across all sites in a zone, not as a spatial gradient (Paper 31).
2. Zone boundaries are sharp, stochastic, Layer I objects – not Allen-Cahn kinks (Paper 28).
3. The half-saturation constant K_{eff} reflects how many copy-forward "votes" are needed to overcome noise, scaling with zone width and diffusion (Papers 29–30).
4. There is an optimal turnover rate ($\nu \approx 0.001$) where zone-level consensus accumulates fastest without being disrupted by excessive births (Papers 30–31).

5 Open Questions for Paper 32

- **Q19.** At what κ exactly does the flat-to-gradient transition occur? Does it coincide with the breakdown of the copy-forward amplification ($\Gamma \to 1$)?
- **Q20.** Does the "vote count" interpretation of zone width explain the sub-linear exponent 0.6? Predict: consensus efficiency $\sim N_{\text{zone}}^{0.6}$ for some model of noisy averaging.
- **Q21.** Can the $\nu = 0.003$ low-amplitude saturation be recovered by increasing FA (stronger consolidation signal)? This would confirm that the disruption is energetic (insufficient consolidation) not structural (broken zone topology).

6 Conclusion

Paper 31 establishes that VCML zones in the standard and lower-diffusion regime are *spatially uniform flat attractors*: within each zone, the field $\mathbf{F}$ has nearly the same value at all columns, with no detectable spatial gradient. This is a direct measurement consequence of the copy-forward mechanism, which averages field values across a zone via the random-walk birth process. Spatial gradients emerge only at high diffusion ($\kappa \geq 0.080$), where they are accompanied by reduced zone differentiation.

The ν-dependence of within-zone structure is absent, confirming that turnover rate affects zone amplitude (not spatial character). The $\nu = 0.003$ failure mode is characterised as low-amplitude saturation: zone structure forms but stabilises at $\approx 40\%$ of the standard amplitude, consistent with excessive birth events disrupting consensus before full consolidation.

These results reframe the Paper 30 scaling: K_{eff} scales with zone width not because of a spatial coherence length, but because wider zones accumulate more "votes" for zone identity, making consensus more efficient. The sub-linear exponent (0.6) reflects saturation of consensus efficiency with vote count.

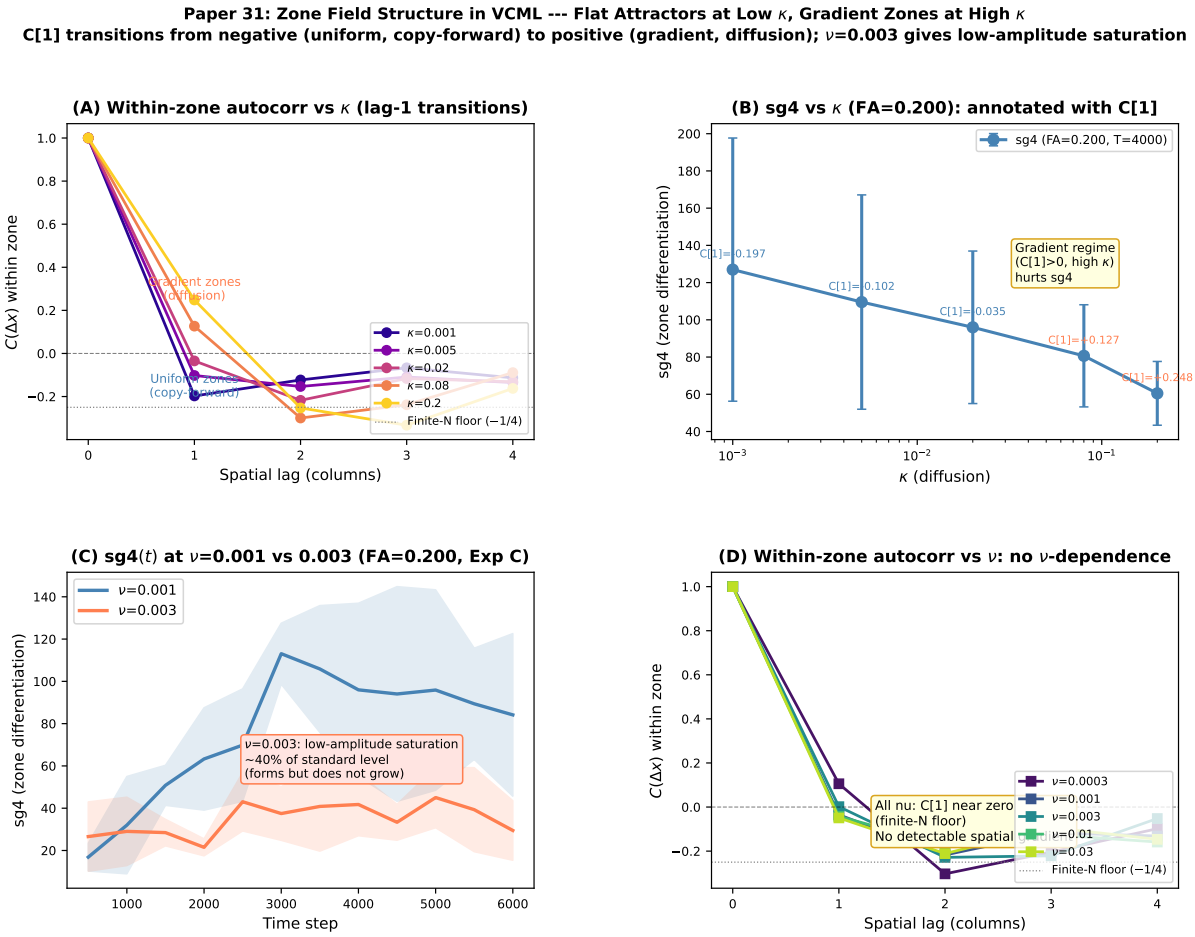

Figure 1: **Paper 31 results.** **(A)** Within-zone autocorrelation $C(\Delta x)$ for Exp A (κ sweep). At low κ (blue/purple), $C[1]$ is near the finite-N floor (dashed line), indicating uniform zones. At high κ (orange/yellow), $C[1] > 0$ indicates genuine spatial gradients. **(B)** sg4 at $T = 4000$ vs κ, annotated with $C[1]$ values. The gradient regime ($C[1] > 0$) corresponds to lower zone differentiation. **(C)** sg4(t) for $\nu = 0.001$ (blue) and $\nu = 0.003$ (red) at FA=0.200. Low-amplitude saturation: structure forms rapidly at $\nu = 0.003$ but stabilises at $\approx 40\%$ of standard amplitude. **(D)** Within-zone autocorrelation for Exp B (ν sweep). No ν-dependence of spatial structure; all values near zero.